STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Thermal equilibrium is transitive.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

Microscopic energy of molecules and their interactions.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

Q=DeltaU+W.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

Q=0.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

Temperature constant.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

No.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

pV=nRT.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

1-T_c/T_h.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

Zero boundary work.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

500 J.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

50%.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

Zero.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

100 J.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Isothermal: T constant; adiabatic: Q=0.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

600 J.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

40%.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

W=pDeltaV=2000 J.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

Second law requires rejected heat or other entropy increase.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

Reversible: ideal quasi-static frictionless; irreversible: friction or finite-temperature heat flow.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

Carnot limit 50%; actual efficiency 33.3%; Qout=1000 J.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.